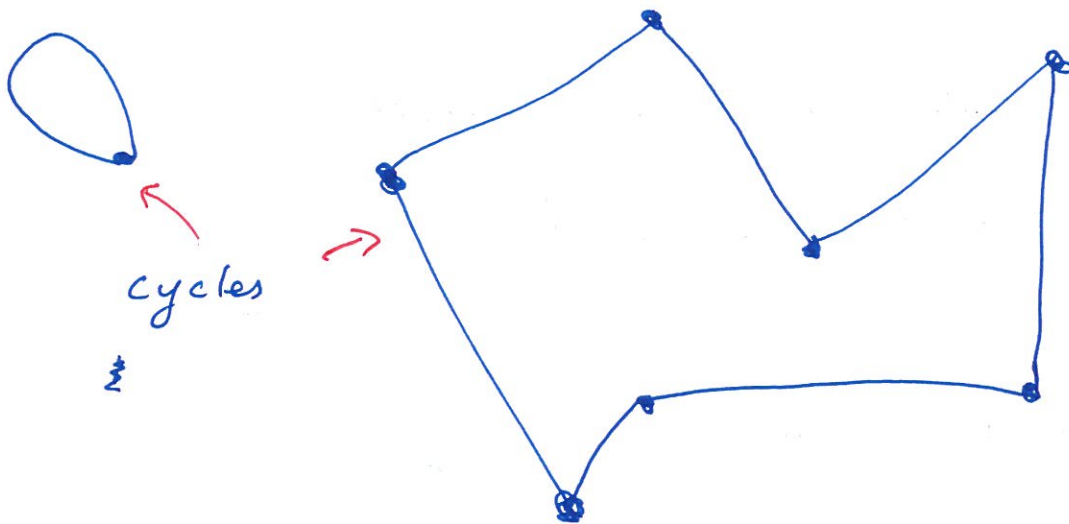


LECTURE 38: MATH1061/7861
TOPICS: TREES
DATE: WEDNESDAY MAY 27, 2026

Learning objectives:

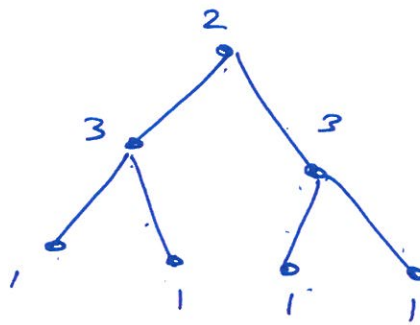
(1) Understand the definition and properties of trees.

A tree is a connected graph
with no cycle.

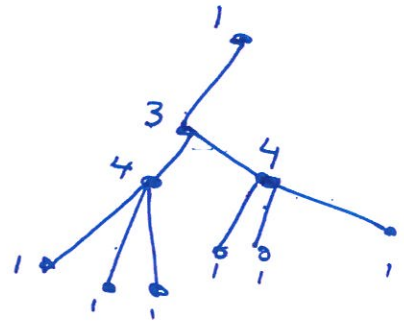




4 vertices
3 edges



7 vertices
6 edges

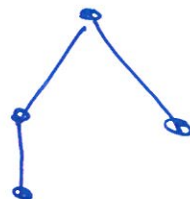


10 vertices
9 edges

* Every ^{is} tree # vertices = # edges + 1

* Every tree with more than one vertex has a vertex of degree 1.

Disconnected:



We assume all graphs are connected.

Q1: Recall a tree is a connected graph which has no circuits (or cycles).

Draw 3 examples of trees. In each case:

Count the number of vertices and edges.

Determine the degree of each vertex.

What do you observe about the relationship between number of vertices and edges?

What do you observe about the smallest degree in each case?

Recall : $\sum \text{degrees} = 2 \times \# \text{ edges.}$
in any graph

(a)

$$2n - 2 = 2(n - 1)$$

Q2: What is the total degree of a tree with n vertices?

(b) A connected graph has 12 vertices and 11 edges. Does it have a cycle? *No. Such a graph must be a tree.*

(c) A graph has 12 vertices and 10 edges. Can it be connected? *No*

Theorem: In a connected graph

$$\# \text{ edges} \geq \# \text{ vertices} - 1$$

And equality holds if and only if the graph is a tree.



Proof: Use induction.

Q3: A tree has two vertices of degree 4, one vertex of degree 2 and all other vertices of degree 1

How many vertices of degree 1 does it have?

Can you draw it?

Let x be the number of vertices of degree 1.

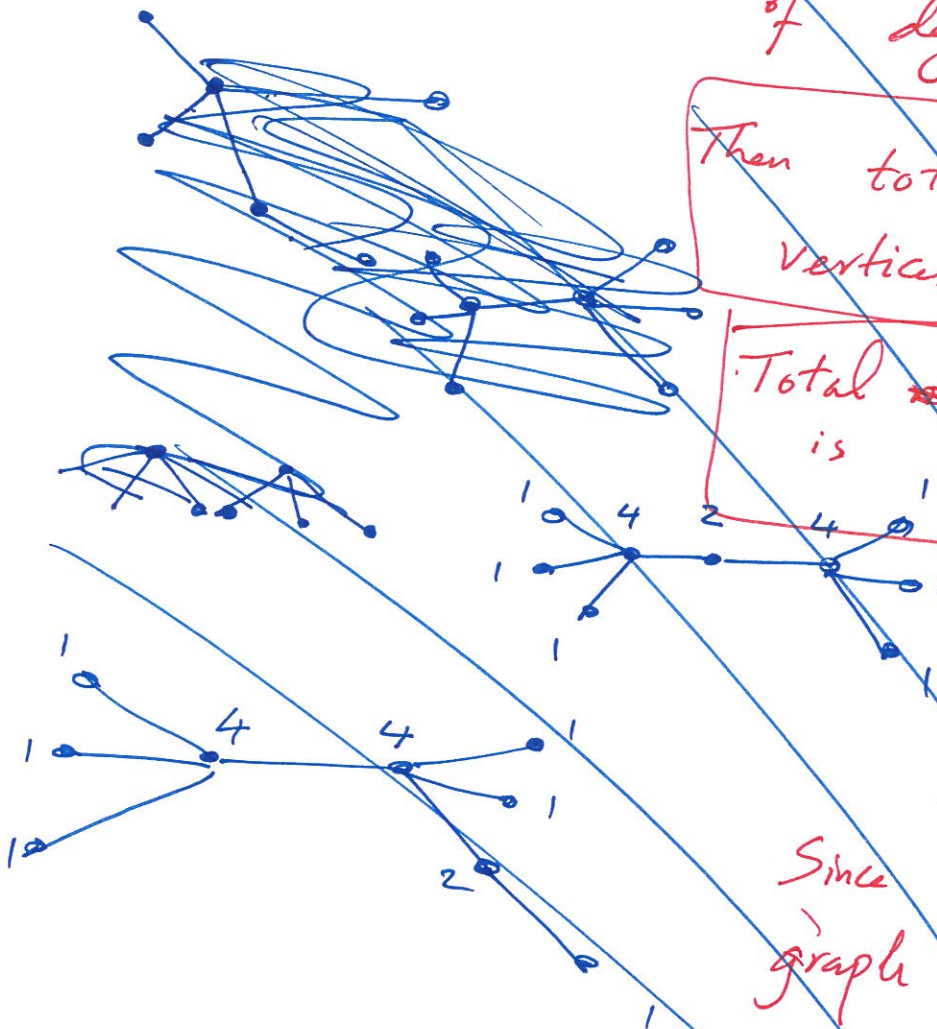
Then total number of vertices is $x + 3$

Total ~~number of~~ degree is $8 + 2 + x$.

Since we know the graph is a tree:

$$\# \text{ edges} = x + 2$$

So we



Let x be the number of vertices of degree 1.

$$\Rightarrow \text{Total number of vertices} = x + 2 + 1 = x + 3$$

Since it's tree

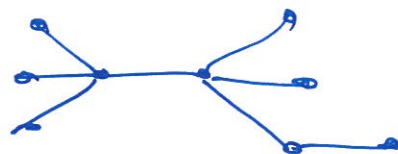
$$\Rightarrow \text{Total number of edges} = x + 2$$

$$\text{Next, total degree} = 8 + 2 + x = x + 10$$

$$\text{So we get: } x + 10 = 2(x + 2)$$

$$\Rightarrow x + 10 = 2x + 4$$

$$\Rightarrow x = 6$$



Remark: A binary search tree is a binary tree where every node satisfies this rule, For each node with value x :

all values in the left subtree are $< x$.

all values in the right subtree are $> x$.

(We assume all values are distinct to keep things simple.)

This is a very important data structure in computer science.

